Vineth Joseph Choudary Mallavarapu

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SUMMARY

M.Sc. Mechatronics and Robotics student at TUM with hands-on experience in Python, deep learning, computer vision, LLM-based interfaces, and robotics software. Experienced in building data-driven perception and automation systems, including image-based classification, autonomous navigation, and synthetic/real-world data pipelines. Interested in applying machine learning and deep learning to vehicle development, simulation, and visual quality assessment.

EDUCATION

Technical University of Munich — Oct 2025 – Present
M.Sc. in Mechatronics, Robotics and Biomechanical Engineering
SRM Institute of Science and Technology — 2020 – 2024
B.Tech. in Mechatronics Engineering with specialization in Robotics — CGPA: 8.16/10

TECHNICAL SKILLS

Programming: Python, C++, MATLAB, TypeScript basics
Machine Learning & AI: Deep learning, computer vision, image classification, LLM-based interfaces, data preprocessing, model evaluation, OpenCV, NumPy, PyTorch, TensorFlow, scikit-learn
Tools & Platforms: Docker, Git, Linux, React, Vite, DroneDeploy
Robotics & Simulation: ROS 2, SLAM, path planning, localization, Gazebo, sensor integration
CAD & Engineering: SolidWorks, ANSYS, AutoCAD

PATENT / PUBLICATION

Patent: System for Identifying and Weighing Loose Items (Millets), App. No. 202541006107 (Published Jan 2025)
Publication: "Development of Automated Millet Identification and Billing System," ADCIS 2024 (Springer)

PROJECTS

Autonomous Mobile Robot using ROS 2 and SLAM — 2025

- Developed an autonomous navigation system using ROS 2 (Jazzy) integrating SLAM, path planning, and control for real-time robot navigation
- Designed ROS 2 nodes for perception, localization, and motion control using Python
- Implemented mapping and navigation pipeline enabling goal-based movement in unknown environments
- Designed modular robotics systems integrating perception, planning, and control components
- Integrated an LLM-based natural language interface to translate high-level user commands into executable robot tasks, connecting AI-based interaction with robotic planning pipelines.

Autonomous Sub-Terrain Exploration with Quadrotor UAV | ROS 2, Unity, OpenCV, OctoMap, Rviz — 2025

- Developed ROS 2 perception modules for an autonomous quadrotor UAV exploring a GPS-denied cave environment in Unity.
- Implemented RGB/depth and semantic-camera lantern detection using OpenCV HSV segmentation, connected components, NMS, depth back-projection, and TF-based world-frame pose estimation.
- Generated RealSense depth-based point clouds and integrated them with OctoMap for 3D voxel-grid mapping and environment reconstruction.
- Integrated and debugged RGB, depth, semantic camera, point-cloud, OctoMap, TF, and object-pose streams using RViz.

Automated Millet Identification and Billing System — 2024

- Developed an image-based **deep learning** classification system for millet variety identification using overhead camera data, achieving 85% classification accuracy after preprocessing and model evaluation.
- Performed dataset preparation, image preprocessing, model testing, and result evaluation to improve classification reliability for real-world billing automation.
- The system autonomously generates a bill based on the measured data.
- The system also includes a touch display with a **Tkinter-based** user interface and an admin portal for updating prices and viewing billing history.

INDUSTRIAL EXPERIENCE

Grace Drone Technologies — Drone Operations Support Engineer — Oct 2024 – Sep 2025

- Assisted in UAV mission planning and execution using DJI Matrice 30T for inspection and mapping tasks.
- Processed aerial data to generate **orthomosaics and site models** using **DroneDeploy** and **AutoCAD**.
- Supported flight operations, telemetry monitoring, and safety validation during real-world deployments.

Fronius India Pvt. Ltd. — Robotics and Welding Automation Intern — May 2024 – Jun 2024

- Conducted experiments on a **FANUC 6-axis industrial robot** to evaluate welding modes **including CMT, PMC, Pulse, and LSC**.
- Analyzed process parameters and weld quality to support manufacturing and process optimization.

LANGUAGE / LEADERSHIP

Languages: English, German (A2)
Powertrain Engineer — Infieon Super Mileage (Shell Eco-Marathon Team): Contributed to energy-efficient propulsion system design within a multidisciplinary team
Association of Mechatronics — Social Service Lead: Coordinated student events and community initiatives